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WEST BENGAL UNIVERSITY OF TECHNOLOGY

IT-604B

COMPUTER GRAPHICS

Time Allotted: 3 Hours | Full Marks: 70

The questions are of equal value.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

GROUP A
(Multiple Choice Type Questions)

1. Answer *all* questions. $10 \times 1 = 10$

(i) User can make any change in image with the use of

(A) non-interactive graphics (B) interactive graphics
(C) both (A) and (B) (D) none of these

(ii) A display controller serves to pass the contents of

(A) frame buffer to monitor (B) monitor to frame buffer
(C) both (A) and (B) (D) none of these

(iii) A 16×16 array of black and white pixels could be represented by _____

(A) 64 bytes (B) 32 bytes
(C) 128 bytes (D) 96 bytes

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(iv) A plotter is capable of

(A) printing a map (B) printing similar images
(C) both (A) and (B) (D) none of these

(v) A graphics tablet works on same principal as

(A) light pen (B) monitor
(C) projector (D) none of these

(vi) In Bresenhan's algorithm error term is initialized to

(A) 0 (B) 1
(C) $-\frac{1}{2}$ (D) none of these

(vii) A line with two endpoint codes as 0000 and 0100 is

(A) partially invisible (B) completely visible
(C) completely invisible (D) trivially invisible

(viii) The best hidden surface removal method used for complex scenes with more than a few thousand surfaces is

(A) scan line algorithm (B) depth buffer algorithm
(C) octree method (D) both (B) and (C)

(ix) $x = at^2$; $y = 2at$ is the parametric equation of

(A) circle (B) rectangular hyperbola
(C) parabola (D) ellipse

(x) The area of computer that is captured by an application is called

(A) window (B) viewport
(C) display (D) none of these

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GROUP B
(Short Answer Type Questions)

Answer any *three* questions. 3×5 = 15

2. Suppose RGB raster system is to be designed using a 8 inch × 10 inch screen with a resolution of 100 pixels per inch in each direction. If we want to store 6 bits per pixel in the frame buffer, how much storage (in bytes) do we need for frame buffer? 5

3. Define persistence in terms of CRT Phosphorous as well as describe different properties of phosphorus and what do you mean by an aspect ratio? 5

4. (a) What do you mean by retracing? Define horizontal as well as vertical retracing. (1+2)+2
   (b) What do you mean by interlacing?

5. (a) What are advantages of DVST over CRT? Also list some disadvantages of DVST. (2+1)+2
   (b) What is the role of a video controller?

6. How much time is spent scanning across each row of pixels during screen refresh on a raster system with resolution of 1280 × 1024 and a refresh rate of 60 frames per second? 5

GROUP C
(Long Answer Type Questions)

Answer any *three* questions. 3×15 = 45

7. (a) Translate the square ABCD whose co-ordinates are A(0, 0), B(3, 0), C(3, 3), D(0, 3) by 2 units in both direction and then scale it by 1.5 units in horizontal direction and 0.5 units in vertical direction. 7+3+5
   (b) Differentiate raster and random scan displays.
   (c) What do you mean by Flood fill algorithm for polygon?

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8. (a) Discuss detail about Cohen-Sutherland line clipping algorithm. 4+3+4+4
   (b) Explain in detail about DDA line drawing algorithm.
   (c) Discuss the Painter's algorithm.
   (d) Describe in detail about Cyrus-Beck clipping algorithms.

9. (a) Explain the significance of Z-buffer algorithm. 5+5+5
   (b) Explain about Bresenham's line drawing algorithm.
   (c) Describe in detail about B-spline curves.

10. (a) What do you mean by window-to-view port co-ordinate transformation? 5+5+5
   (b) Explain in detail about video display devices with neat diagram.
   (c) Discuss in detail about Bresenham's circle generating algorithms.

11. Write short notes on any *three* of the following: 3×5
   (a) Rotation Transformation
   (b) Anti-aliasing
   (c) Shearing Transformation
   (d) Reflection Transformation
   (e) Refresh buffer.